

# Lingze Zeng

lingze.zone | lingze@comp.nus.edu.sg | zlz.bit.pro@gmail.com | +65-88426606

Available for internship/research opportunities: **Now – Aug 2027** (Expected Graduation)

## EDUCATION

|                                                                                                                                 |                                                       |
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| <b>National University of Singapore (NUS)</b><br><i>Ph.D. in Computer Science, School of Computing</i>                          | <b>2023 – Present</b><br>Advisor: Prof. Ooi Beng Chin |
| <b>Beijing Institute of Technology (BIT)</b><br><i>B.Sc. in Software Engineering, School of Computer Science and Technology</i> | <b>2016 – 2020</b><br>Advisor: Prof. Zhang Meihui     |

## PUBLICATIONS

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| <u>pTNAS: Progressive Neural Architecture Search for Tabular Data</u><br>Naili Xing, Shaofeng Cai, <b>Lingze Zeng</b> , Jiaqi Zhu, Peng Lu, Jian Pei, Beng Chin Ooi.                                                 | <b>ICML 2026</b>          |
| <u>NRBench: A Benchmark Suite for Learned Database Components with Drift Modeling</u><br>Zhanhao Zhao, Haotian Gao, Naili Xing, <b>Lingze Zeng</b> , Meihui Zhang, Gang Chen, Manuel Rigger, Beng Chin Ooi.          | <b>SIGMOD 2026</b>        |
| <u>From Schema to Signal: Retrieval-Augmented Modeling for Relational Data Analytics</u><br><b>Lingze Zeng</b> , Shaofeng Cai, Changshuo Liu, Zhongle Xie, Yuncheng Wu, Beng Chin Ooi                                | <b>ArXiv 2025</b>         |
| <u>NeurIDA: Dynamic Modeling for Effective In-Database Analytics</u><br><b>Lingze Zeng</b> , Naili Xing, Shaofeng Cai, Peng Lu, Gang Chen, Jian Pei, Beng Chin Ooi.                                                  | <b>ArXiv 2025</b>         |
| <u>NeuralCohort: Cohort-aware Neural Representation Learning for Healthcare Analytics</u> [pdf]<br>Changshuo Liu, <b>Lingze Zeng</b> , Kaiping Zheng, Shaofeng Cai, Beng Chin Ooi, James Wei Luen Yip.               | <b>ICML 2025</b>          |
| <u>Powering In-database Dynamic Model Slicing for Structured Data Analytics</u> [pdf]<br><b>Lingze Zeng</b> , Naili Xing, Shaofeng Cai, Gang Chen, Beng Chin Ooi, Jian Pei, Yuncheng Wu.                             | <b>VLDB 2025</b>          |
| <u>NeurDB: An AI-powered Autonomous Data System</u> [pdf]<br>Beng Chin Ooi, Shaofeng Cai, Gang Chen, Kian Lee Tan, Yuncheng Wu, Xiaokui Xiao, Naili Xing, Cong Yue, <b>Lingze Zeng</b> , Meihui Zhang, Zhanhao Zhao. | <b>Science China 2024</b> |
| <u>Multi-label Classification with Long-tailed Distribution and Partial Labels</u> [pdf]<br>Wenqiao Zhang, Changshuo Liu, <b>Lingze Zeng</b> , Beng Chin Ooi, Siliang Tang, Yuncheng Wu.                             | <b>ICCV 2023</b>          |
| <u>A Two-phase Approach for Recognizing Tables with Complex Structures</u> [link]<br>Huichao Li, <b>Lingze Zeng</b> , Weiyu Zhang, Jianing Zhang, Ju Fan, Meihui Zhang.                                              | <b>DASFAA 2022</b>        |
| <u>Chinese Synonym Discovery on Privacy-constrained Medical Terms with Pre-trained BERT</u><br><b>Lingze Zeng</b> , Chang Yao, Meihui Zhang, Zhongle Xie.                                                            | <b>APWeb 2022</b>         |

## RESEARCH EXPERIENCE

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| <b>Research Assistant</b><br><i>National University of Singapore (NUS)</i> <ul style="list-style-type: none"><li>Topics: <i>Data Engine for Cohort Analysis</i>; <i>Relational Data Analytics</i></li><li>Contributed to the Cohort OLAP system (<a href="#">COOL</a>)</li></ul> | <b>2022 – 2023</b> |
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**Research Assistant**

2020 – 2021

*Institute of Computing Innovation, Zhejiang University (ZJU)*

- Topics: *Named Entity Recognition; Medical Synonym Discovery*
- CHIP 2021 Competition Task 1 — **Rank 1st** solution
- Contributed to NLP open-source tool [ark-nlp](#)

TEACHING EXPERIENCE

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**Teaching Assistant**, *National University of Singapore (NUS)*

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|-------------------------------------------------------------------------------------------------|-----------|
| • CP2107 Summer Project — <i>In-Database Feature Store for Lightweight Analytical Databases</i> | 2025–2026 |
| • CP2107 Summer Project — <i>Tabular Foundation Model for Data Analytics</i>                    | 2025–2026 |
| • CS4221 Database Application and Tuning                                                        | 2024–2025 |
| • CS5344 Big-Data Analytics Technology                                                          | 2023–2024 |

AWARDS

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|-------------------------------------------------------------------------|------------|
| • SIGMOD Systems Award 2024 – <i>Apach Singa</i>                        | 2024       |
| • Research Achievement Award – <i>National University of Singapore</i>  | 2023       |
| • Outstanding Graduate Student – <i>Beijing Institute of Technology</i> | 2020       |
| • JJWorld Scholarship – <i>Beijing Institute of Technology</i>          | 2018       |
| • First-class Scholarship, – <i>Beijing Institute of Technology</i>     | 2017, 2019 |

RESEARCH INTERESTS

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- Unified analytics over structured and relational data.
- AIxDB systems; In-database analytics.
- Application and extension of tabular foundation model

SKILL STACK

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- **Languages:** Python, C, C++, Java, Go, Markdown, SQL
- **Vibe Coding:** Claude Code, Codex, Copilot
- **Teamwork:** Slack, Git